AVIM — Artifact Version Integrity Module

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OOF® Origin Open Foundation®

Independent Methodological Authority

Architecture: SLGA™ — Software Lifecycle Governance Architecture

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Category: Governance & Enforcement

Subcategory: Software Lifecycle Governance

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Governed Space: Software Artifact Version Integrity Governance

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Canonical Definition

The Artifact Version Integrity Module (AVIM) governs the identification,

version attribution, content correspondence, integrity verification, evidence preservation, and lifecycle continuity of software artifacts and their version relationships.

AVIM establishes the methodological conditions under which a software

artifact can be recognized as an identifiable, version-associated, evidence-supported object whose claimed identity and version remain distinguishable from its actual content, recorded provenance, and observed integrity state.

It governs the relationships among artifact identity, artifact content,

version identifiers, build references, configuration baselines, artifact transformations, repository records, integrity evidence, and subsequent artifact states.

AVIM distinguishes artifact naming from artifact identity, version labeling

from verified version correspondence, content integrity from software correctness, and an artifact's recorded origin from independently supported provenance.

It applies to software artifacts created, transformed, stored, copied,

packaged, or referenced by humans, organizations, automated pipelines, AI coding assistants, autonomous development agents, and software infrastructure systems.

Its principal output is:

Governed Artifact Version Integrity State

Its foundational distinction is:

Declared Artifact Version ≠ Verified Artifact Version Integrity.

Operational Role

The operational role of AVIM is:

To ensure that material software artifacts remain identifiable, correctly associated with their claimed versions, supported by competent integrity and correspondence evidence, and traceable through relevant transformations and lifecycle transitions without confusing artifact integrity with software correctness, security, or release authorization.

AVIM provides the artifact-version integrity governance mechanism of SCFS.

It establishes whether an artifact's identity, content, version reference,

and material lifecycle relationships can be sufficiently supported.

AVIM does not independently authorize software creation, execute builds, approve configuration changes, validate software quality, certify security, or authorize release and deployment.

It provides an evidence-supported artifact-version integrity state for

competent downstream SCFS and SLGA™ governance.

Operational Question

How can a software artifact's identity, claimed version, actual content, build relationship, and lifecycle continuity remain sufficiently established and verifiable so that a version label or repository record is not mistaken for proof of artifact integrity?

Position in Architecture Governance Flow

The SLGA™ governance flow is:

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting:

Software Lifecycle Control

AVIM operates within:

Software Configuration Standard (SCFS)

The SCFS internal governance flow is:

Configuration Baseline → Build Management → Artifact Version Integrity →

Configuration Change Control → Build Reproducibility & Environment

AVIM governs:

Artifact Version Integrity

It is the third Core Module of SCFS.

Its principal upstream references are:

Governed Configuration Baseline State

from:

CBM — Configuration Baseline Module

and:

Governed Software Build State

from:

BMM — Build Management Module

Its principal output is:

Governed Artifact Version Integrity State

This output supports:

CCCM — Configuration Change Control Module

BREM — Build Reproducibility & Environment Module

and contributes to:

Governed Software Configuration State

AVIM may be invoked whenever an artifact is created, transformed, versioned, registered, copied, promoted, retrieved, replaced, or reassessed.

• Governed Space

Software Artifact Version Integrity Governance

This space governs the conditions under which software artifacts and their

version relationships remain identifiable and evidence-supported.

Its principal dimensions include:

• Artifact governance object identity

• Artifact purpose

• Artifact type

• Artifact scope

• Artifact classification

• Artifact identity

• Artifact content identity

• Artifact version identity

• Version attribution

• Version correspondence

• Artifact composition references

• Configuration baseline correspondence

• Build correspondence

• Artifact provenance

• Artifact transformation relationships

• Artifact derivation

• Artifact equivalence

• Artifact duplication

• Artifact storage references

• Artifact repository records

• Artifact retrieval

• Artifact integrity evidence

• Integrity verification

• Verification method suitability

• Verification authority

• Artifact modification

• Artifact replacement

• Artifact promotion references

• Artifact distribution references

• Artifact version conflicts

• Artifact integrity uncertainty

• Artifact evidence gaps

• Critical artifact integrity gaps

• Artifact continuity

• Artifact reassessment

• Artifact governance states

The governed space is independent of any particular artifact format, programming language, repository platform, build system, hashing algorithm, or software delivery technology.

Software Artifact Governance Object

Every material AVIM assessment shall establish a Software Artifact

Governance Object.

The object represents an identifiable software artifact and the governance

conditions under which its version integrity may be assessed.

It should identify:

Artifact Identity: The software artifact being assessed.

Artifact Purpose: The artifact's intended function.

Artifact Type: The recognized artifact form.

Artifact Scope: The boundaries of the assessed artifact.

Artifact Classification: The artifact's intended lifecycle role.

Content Identity: The identifiable artifact content.

Version Identity: The claimed or established version.

Build Reference: The relevant build relationship.

Baseline Reference: The applicable configuration baseline relationship.

Provenance Reference: The recorded artifact origin and transformation

history.

Integrity Evidence: Evidence supporting artifact integrity.

Verification State: The condition of integrity assessment.

Lifecycle References: Relevant artifact transitions.

Governance State: The recognized artifact-version integrity condition.

The governance object shall remain distinguishable from a filename, version

label, repository entry, or checksum.

Artifact Governance Object ≠ Artifact Filename.

Artifact Purpose Governance

AVIM shall identify the purpose for which the artifact exists.

Purposes may include:

• Development

• Testing

• Integration

• Distribution

• Release preparation

• Deployment

• Maintenance

• Recovery

• Historical preservation

Artifact purpose shall remain distinguishable from its technical properties

and authorization status.

Artifact Purpose ≠ Artifact Authorization.

Artifact Type Governance

AVIM shall identify the relevant artifact type.

Types may include:

• Executable binaries

• Libraries

• Software packages

• Container images

• Compiled assets

• Generated code

• Deployable bundles

• Configuration packages

• Firmware images

• Software installation packages

Artifact type shall remain sufficiently precise for competent identity and

integrity assessment.

Artifact Type ≠ Artifact Identity.

Artifact Scope Governance

AVIM shall establish the boundaries of the artifact being assessed.

Scope may concern:

• A single file

• A software package

• A container image

• A multi-file bundle

• A composite artifact

• A generated software component

• A deployable software unit

Scope shall identify which constituent elements are included in the integrity assessment.

Artifact Scope Definition ≠ Verified Artifact Composition.

Artifact Classification

AVIM shall recognize the artifact's intended lifecycle classification.

Classifications may include:

• Development artifact

• Test artifact

• Integration artifact

• Release candidate

• Production artifact

• Maintenance artifact

• Recovery artifact

• Historical artifact

Classification shall not independently establish artifact validity or

release authorization.

Artifact Classification ≠ Artifact Governance State.

Artifact Identity Governance

AVIM shall establish an identifiable reference for each material artifact.

Identity may involve:

• Artifact identifier

• Artifact namespace

• Artifact type

• Artifact content reference

• Artifact creation reference

• Artifact repository reference

• Relevant lifecycle context

An artifact's identity shall remain distinguishable from its human-readable

name.

Artifact Name ≠ Verified Artifact Identity.

Artifact Content Identity

AVIM shall establish the identity of the artifact's material content.

Content identity may be supported through:

• Cryptographic digests

• Content-addressed references

• Signed manifests

• Verified package records

• Direct artifact inspection

• Competent equivalent mechanisms

The selected method shall be appropriate to the artifact type and integrity

claim.

Artifact Label ≠ Artifact Content Identity.

Artifact Version Identity

AVIM shall identify the version attributed to a material artifact.

Version references may include:

• Semantic versions

• Build versions

• Release candidate identifiers

• Internal revision identifiers

• Package versions

• Container tags

• Artifact generation identifiers

Version identity shall remain associated with the artifact to which it

applies.

Version Identifier ≠ Verified Artifact Version.

Version Attribution Governance

AVIM shall preserve the basis upon which a version is attributed to an artifact.

Attribution may originate from:

• Build records

• Version manifests

• Repository metadata

• Package metadata

• Release preparation records

• Authorized version assignment

Attribution shall remain distinguishable from independent content

verification.

Version Attribution Record ≠ Version Correspondence Evidence.

Version Correspondence

AVIM shall assess whether an artifact corresponds to its claimed version.

Correspondence may involve:

• Claimed version

• Expected artifact identity

• Expected content reference

• Build identity

• Applicable baseline

• Version assignment evidence

• Observed artifact content

Version correspondence shall be evaluated within the relevant artifact scope.

Matching Version Label ≠ Verified Version Correspondence.

Artifact Composition References

AVIM shall identify material constituent elements of composite artifacts

where necessary for integrity assessment.

Composition may include:

• Files

• Libraries

• Embedded components

• Generated resources

• Package metadata

• Container layers

• Configuration resources

AVIM governs the integrity and identity relationships of the resulting

artifact.

Dependency origin and suitability remain governed by SDS.

Composite Artifact Identity ≠ Verified Constituent Integrity.

Configuration Baseline Correspondence

AVIM shall preserve the relationship between an artifact and its relevant

configuration baseline.

The relationship may identify:

• Baseline identity

• Expected artifact reference

• Relevant version constraints

• Material configuration relationships

• Applicable environment assumptions

Baseline correspondence shall remain distinguishable from actual runtime

configuration conformity.

Artifact Baseline Correspondence ≠ Verified Runtime Configuration.

Build Correspondence

AVIM shall preserve evidence-supported relationships between artifacts and

their originating build events.

Relevant information may include:

• Build identity

• Build output reference

• Source input references

• Toolchain references

• Relevant execution conditions

• Artifact creation evidence

BMM governs the build execution process.

AVIM governs whether the resulting artifact identity and version

relationship remain sufficiently supported.

Build Reference ≠ Verified Artifact Build Origin.

Artifact Provenance

AVIM shall preserve material evidence concerning artifact origin and

relevant transformations.

Provenance may involve:

• Originating build

• Producing system

• Source references

• Artifact transformation events

• Repository transfers

• Relevant signing events

• Custody relationships

A recorded provenance claim shall remain distinguishable from independently

verified provenance.

Recorded Artifact Provenance ≠ Verified Artifact Provenance.

Artifact Transformation Governance

AVIM shall identify material transformations that may affect artifact identity or content.

Transformations may include:

• Packaging

• Compression

• Re-signing

• Rebundling

• Recompilation

• Container layering

• Metadata modification

• Binary patching

Transformations shall preserve the relationship between predecessor and

successor artifacts.

Artifact Transformation ≠ Artifact Identity Preservation.

Artifact Derivation

AVIM shall establish material relationships between derived artifacts and

their source artifacts.

Derivation may involve:

• Parent artifact identity

• Transformation method

• Transformation event

• Derived artifact identity

• Resulting content reference

• Supporting evidence

Derived artifacts shall not automatically inherit the integrity status of

their predecessors.

Verified Source Artifact ≠ Verified Derived Artifact.

Artifact Equivalence

AVIM shall distinguish different forms of artifact equivalence.

Equivalence may concern:

• Identical content

• Identical package composition

• Identical executable behavior

• Equivalent functional output

• Equivalent deployment suitability

These claims require different evidence.

AVIM governs artifact identity and content correspondence.

Broader functional equivalence remains subject to competent software

quality governance.

Functional Equivalence ≠ Content Identity.

Artifact Duplication

AVIM shall recognize when multiple artifact instances claim the same identity or version.

Instances may exist in:

• Different repositories

• Build systems

• Artifact stores

• Deployment environments

• Backup systems

• Distribution channels

Matching labels shall not establish identical content.

Duplicate Version Labels ≠ Identical Artifact Content.

Artifact Storage References

AVIM shall preserve material storage references relevant to artifact

integrity and retrieval.

References may include:

• Repository location

• Artifact store identity

• Storage object reference

• Content address

• Relevant access conditions

• Retention context

Storage references shall remain distinguishable from verified artifact

content.

Artifact Storage Location ≠ Artifact Identity.

Artifact Repository Governance

AVIM shall assess material repository records supporting artifact identity

and version claims.

Relevant records may include:

• Artifact registration

• Version assignment

• Content digest

• Upload event

• Modification event

• Repository promotion

• Deletion or replacement event

Repository records shall remain subject to evidence integrity assessment.

Repository Record ≠ Independent Artifact Verification.

Artifact Retrieval Governance

AVIM shall recognize the artifact actually retrieved from a storage or

distribution source.

Retrieval assessment may involve:

• Requested artifact identity

• Requested version

• Source location

• Retrieved content

• Observed digest

• Retrieval time

• Verification result

The retrieved artifact shall not automatically be treated as identical to

the intended artifact.

Requested Artifact ≠ Retrieved Artifact.

Artifact Integrity Evidence

AVIM shall preserve evidence supporting material artifact integrity claims.

Evidence may include:

• Cryptographic digests

• Digital signatures

• Signed manifests

• Verified build references

• Repository integrity records

• Artifact inspection results

• Relevant custody records

Evidence shall remain associated with the specific artifact and integrity

claim it supports.

Integrity Evidence Presence ≠ Verified Artifact Integrity.

Integrity Verification

AVIM shall establish whether the material artifact integrity claim has been

sufficiently verified.

Verification may assess:

• Artifact content identity

• Expected digest correspondence

• Signature validity

• Manifest correspondence

• Version reference correspondence

• Artifact transformation history

• Relevant evidence integrity

Verification shall remain bounded by the method and evidence used.

Successful Integrity Check ≠ Complete Artifact Trustworthiness.

Verification Method Suitability

AVIM shall assess whether the selected verification method is appropriate for the relevant artifact and claim.

Relevant considerations may include:

• Artifact type

• Content scope

• Verification mechanism

• Algorithm suitability

• Reference authenticity

• Key or credential validity

• Material threat conditions

A technically successful verification using an unsuitable or untrusted

reference may not establish the intended integrity claim.

Verification Mechanism Execution ≠ Competent Integrity Verification.

Integrity Reference Authenticity

AVIM shall distinguish the integrity of an artifact from the authenticity

of the reference against which it is checked.

For example:

A downloaded artifact may match a published checksum.

However, if the checksum was obtained from an untrusted or compromised

source, the match does not independently establish that the artifact is the intended authorized version.

Digest Match ≠ Trusted Artifact Origin.

Artifact Signing References

AVIM shall preserve relevant signing information where signatures are used to support artifact integrity or provenance claims.

Relevant information may include:

• Signing identity

• Signing authority

• Signed artifact scope

• Signature validity

• Signing time

• Certificate or key status

• Applicable trust conditions

A valid signature does not independently establish software correctness or legitimate release approval.

Valid Artifact Signature ≠ Authorized Software Release.

Verification Authority

AVIM shall identify the competent authority or accepted mechanism

responsible for material integrity determinations.

Authority may concern:

• Approved verification procedures

• Trusted reference sources

• Assigned artifact integrity responsibilities

• Defined organizational controls

• Applicable automated verification mechanisms

Technical ability to perform verification shall remain distinguishable from

authority to certify or accept its result.

Verification Capability ≠ Verification Authority.

Artifact Modification Recognition

AVIM shall identify material changes affecting artifact content or identity.

Modifications may include:

• Binary patching

• Metadata alteration

• Package modification

• File replacement

• Container layer changes

• Signature replacement

• Configuration resource modification

A modified artifact shall be reassessed where its previous integrity determination no longer applies.

Previous Artifact Integrity ≠ Current Artifact Integrity.

Artifact Replacement

AVIM shall preserve the relationship between an existing artifact and its

replacement.

Replacement should identify:

• Previous artifact

• Replacement artifact

• Material content differences

• Version relationship

• Replacement event

• Relevant authority reference

• Supporting evidence

Artifact replacement shall not silently preserve the previous artifact's

identity or integrity state.

Artifact Replacement ≠ Artifact Identity Continuity.

Artifact Promotion References

AVIM shall recognize material transitions in which an artifact is promoted between lifecycle contexts.

Examples may include:

• Development to testing

• Testing to staging

• Staging to release preparation

• Release preparation to production artifact storage

AVIM preserves artifact identity and integrity across these transitions.

Authorization of release and deployment remains governed by the relevant

SLGA™ standards.

Artifact Promotion ≠ Release Authorization.

Artifact Distribution References

AVIM shall preserve material integrity relationships when artifacts are distributed or transferred.

Relevant information may include:

• Origin artifact

• Distribution source

• Transfer event

• Destination reference

• Received artifact

• Content verification

• Relevant custody evidence

Distribution shall not automatically establish that the recipient obtained the intended artifact.

Artifact Distribution ≠ Verified Artifact Receipt.

Artifact Version Conflict

An Artifact Version Conflict exists where material version claims or supporting evidence are incompatible.

Examples may include:

• One version label associated with different artifact content

• Conflicting version manifests

• Contradictory repository metadata

• Incompatible build-to-version references

• Conflicting artifact signing information

AVIM shall preserve the conflict and its supporting evidence.

Conflicting Version Records ≠ Permission to Select the Preferred Version.

Artifact Identity Conflict

An Artifact Identity Conflict exists where material claims concerning artifact identity or content correspondence are incompatible.

Examples may include:

• Identical artifact identifiers with different content

• Conflicting content digests

• Inconsistent package composition

• Contradictory provenance records

• Different retrieved content under the same repository reference

AVIM shall preserve the conflict for competent assessment.

Shared Artifact Identifier ≠ Shared Artifact Identity.

Artifact Integrity Uncertainty

AVIM shall preserve material uncertainty concerning artifact identity,

version correspondence, provenance, or integrity.

Uncertainty may arise where:

• Artifact content cannot be inspected

• Expected digests are unavailable

• Version attribution is ambiguous

• Build origin is unresolved

• Signing references cannot be validated

• Transformation history is incomplete

• Retrieved content differs from expectations

Uncertainty shall not be silently converted into verified integrity.

Unresolved Artifact Integrity ≠ Verified Artifact Integrity.

Artifact Evidence Gap

An Artifact Evidence Gap exists where material information required for competent artifact-version integrity assessment remains insufficiently established.

Examples may include:

• Missing content identity evidence

• Missing version attribution records

• Unresolved build correspondence

• Missing transformation evidence

• Incomplete repository records

• Unverified integrity references

AVIM shall identify the gap and its material significance.

Artifact Evidence Gap ≠ Proven Artifact Compromise.

Critical Artifact Integrity Gap

A Critical Artifact Integrity Gap exists where missing evidence prevents competent assessment of a materially consequential artifact.

Examples may include:

- A production executable without sufficiently established content

identity. - A release candidate whose version cannot be reliably connected to its build. - A critical software package with unresolved signature authenticity. - A deployment artifact whose retrieved content cannot be matched to its approved reference.

AVIM shall preserve the critical gap for competent downstream assessment.

It does not independently determine software compromise, operational shutdown, or legal liability.

Critical Artifact Integrity Gap ≠ Automatic Artifact Compromise.

Artifact Integrity Failure

An Artifact Integrity Failure exists where competent evidence establishes that a material artifact integrity requirement is not satisfied.

Examples may include:

• A verified digest mismatch

• An invalid signature where signature validity is required

• A demonstrated artifact-version mismatch

• A confirmed unauthorized content alteration within the assessed

integrity scope.

AVIM shall distinguish established integrity failure from uncertainty or

missing evidence.

Verified Integrity Failure ≠ Mere Integrity Uncertainty.

Artifact Integrity Validation

AVIM shall determine whether material artifact-version integrity conditions

are sufficiently supported.

Validation may consider:

• Artifact identity

• Content identity

• Version attribution

• Version correspondence

• Build correspondence

• Baseline correspondence

• Provenance

• Transformation relationships

• Integrity evidence

• Verification method suitability

• Material uncertainty

Validation shall remain limited to the artifact properties and claims

actually assessed.

Validated Artifact Integrity ≠ Validated Software Correctness.

Artifact Continuity

AVIM shall preserve material artifact identity and version relationships

across software lifecycle transitions.

Continuity may involve:

• Artifact creation

• Build output registration

• Repository storage

• Artifact copying

• Artifact transformation

• Version assignment

• Promotion

• Distribution

• Retrieval

• Replacement

• Historical preservation

Artifact continuity shall distinguish identical content from derived,

transformed, or replaced content.

Artifact Continuity ≠ Artifact Content Immutability.

Artifact Historical Preservation

AVIM shall support preservation of material historical artifact identity

and version evidence.

Historical records may include:

• Previous artifact identities

• Previous content references

• Previous version assignments

• Superseded artifact references

• Relevant integrity assessments

• Transformation histories

• Replacement relationships

Historical preservation shall not automatically establish that a historical artifact remains currently suitable for use.

Historical Artifact Integrity ≠ Current Artifact Suitability.

Artifact Reassessment

AVIM shall support reassessment when material artifact identity, version, provenance, or integrity conditions change.

Triggers may include:

• New artifact content evidence

• Conflicting repository records

• Signature invalidation

• Discovered version mismatch

• Artifact modification

• Artifact replacement

• New provenance evidence

• Material integrity gaps

• Changed verification references

Reassessment shall preserve relevant historical determinations and their

supporting evidence.

Artifact Reassessment ≠ Historical Integrity Record Erasure.

Artifact Governance States

AVIM shall support identifiable governance states.

Possible states include:

Identified: An artifact governance object has been recognized.

Registered: An artifact has been associated with a material record.

Version-Declared: A version claim has been established as a declaration.

Under Assessment: Material artifact integrity conditions are being evaluated.

Content-Verified: Artifact content identity has been sufficiently

established within the assessed scope.

Version-Verified: Artifact-version correspondence has been sufficiently

established.

Provenance-Unresolved: Material artifact origin evidence remains

insufficient.

Version-Unresolved: Material version correspondence remains uncertain.

Integrity-Unresolved: Material integrity conditions remain insufficiently

established.

Evidence-Limited: Available evidence is insufficient for the required determination.

Conflicted: Material artifact claims or evidence are contradictory.

Integrity-Failed: A material integrity requirement has been shown not to

be satisfied.

Validated: Material artifact-version integrity conditions are sufficiently supported within the assessed scope.

Superseded: A later artifact or version relationship has replaced the

earlier reference.

Retired: The artifact is no longer an active applicable reference.

Unresolved: Available information does not support a sufficiently determined state.

These states shall not independently establish software correctness,

security, release readiness, or deployment authorization.

Verified Artifact Integrity ≠ Verified Software Safety.

Governed Artifact Version Integrity State

The principal output of AVIM is:

Governed Artifact Version Integrity State

The state shall preserve sufficient information to identify:

Artifact Governance Object: The artifact assessment.

Artifact Identity: The identifiable software artifact.

Artifact Purpose: The intended function.

Artifact Type: The recognized artifact form.

Artifact Scope: The boundaries of the assessed artifact.

Artifact Classification: The intended lifecycle role.

Content Identity: The established or claimed artifact content identity.

Version Identity: The claimed or established version.

Version Attribution: The basis for version assignment.

Version Correspondence: The assessed relationship between version claim

and artifact content.

Composition References: Material constituent artifact references.

Baseline Correspondence: The relevant configuration baseline relationship.

Build Correspondence: The relevant build-to-artifact relationship.

Artifact Provenance: The recorded and assessed origin relationships.

Transformation Relationships: Material artifact transformations.

Derivation Relationships: Relevant predecessor and successor artifacts.

Equivalence Claims: Material equivalence claims and their evidence.

Duplication References: Relevant artifact instance relationships.

Storage References: Material artifact storage locations.

Repository Records: Relevant repository identity and version evidence.

Retrieval Evidence: The relationship between requested and retrieved

artifacts.

Integrity Evidence: Evidence supporting material integrity claims.

Verification Results: The results of competent integrity checks.

Verification Method: The mechanism used to assess integrity.

Reference Authenticity: The trust condition of integrity references.

Signing References: Relevant artifact signature information.

Verification Authority: The competent basis for integrity determinations.

Modification History: Material content or identity changes.

Replacement Relationships: Relevant artifact replacement records.

Promotion References: Relevant lifecycle promotion events.

Distribution References: Relevant transfer and receipt relationships.

Version Conflicts: Contradictory version claims.

Identity Conflicts: Contradictory artifact identity claims.

Integrity Uncertainty: Material unresolved conditions.

Evidence Gaps: Missing material artifact evidence.

Critical Integrity Gaps: Material gaps requiring competent attention.

Integrity Failures: Established integrity failures.

Integrity Validation: The scope and result of artifact integrity

assessment.

Artifact Continuity: Preservation of material lifecycle relationships.

Historical Preservation: Relevant historical artifact records.

Artifact Reassessment: Material reassessment history.

Artifact Governance State: The recognized artifact-version integrity

condition.

Downstream References: Information required by competent SCFS and SLGA™

governance.

The state shall not independently establish software quality, security,

release authorization, deployment authorization, or build reproducibility.

Minimum Implementation Framework

A conforming implementation of AVIM must provide the following six functional capabilities.

• Artifact Object, Identity, Purpose, Type, Scope, and Classification

Governance

Capability to establish identifiable artifact governance objects, determine

artifact purpose, type, scope, and classification, and distinguish artifact identity from filenames, repository references, and informal labels.

The implementation must preserve sufficiently stable artifact identity across relevant lifecycle events.

• Artifact Content, Version Attribution, and Correspondence Governance

Capability to establish artifact content identity, preserve version

attribution, assess artifact-version correspondence, and distinguish declared version labels from verified content relationships.

The implementation must not treat matching version labels as proof of identical artifact content.

• Build, Baseline, Provenance, Transformation, and Derivation Governance

Capability to associate artifacts with relevant build and configuration

baseline references, preserve material provenance evidence, identify transformations and derivations, and maintain predecessor–successor relationships.

The implementation must distinguish recorded provenance from verified provenance and original artifacts from transformed artifacts.

• Artifact Integrity Evidence, Verification, Reference Authenticity, and

Authority Governance

Capability to associate artifact integrity claims with competent evidence,

apply suitable verification methods, assess reference authenticity, preserve signing information where relevant, and identify competent verification authority.

The implementation must distinguish successful technical checks from complete artifact trustworthiness.

• Artifact Conflict, Gap, Failure, Continuity, and Reassessment Governance

Capability to identify version conflicts, identity conflicts, integrity

uncertainty, evidence gaps, critical integrity gaps, established integrity failures, modification events, replacement relationships, historical records, and reassessment conditions.

The implementation must distinguish missing evidence from established compromise or integrity failure.

• Governed Artifact Version Integrity State and SCFS Handoff Governance

Capability to produce a Governed Artifact Version Integrity State with

sufficient artifact identity, version, content, provenance, build correspondence, integrity evidence, validation, uncertainty, and continuity information for competent downstream SCFS governance.

The implementation must not represent artifact integrity as proof of software correctness, security, release readiness, or deployment authorization.

GIE™ Compatibility

AVIM is compatible with GIE™ as a downstream methodological analysis and execution layer.

GIE™ may support:

• Artifact governance object identification

• Artifact purpose analysis

• Artifact type classification

• Artifact scope assessment

• Artifact identity analysis

• Content identity reference analysis

• Version identity analysis

• Version attribution assessment

• Version correspondence analysis

• Artifact composition mapping

• Baseline correspondence analysis

• Build correspondence analysis

• Artifact provenance mapping

• Transformation relationship analysis

• Artifact derivation mapping

• Equivalence claim classification

• Artifact duplication detection

• Storage reference analysis

• Repository record assessment

• Artifact retrieval correspondence analysis

• Integrity evidence association

• Integrity verification support

• Verification method suitability analysis

• Integrity reference authenticity assessment

• Signing reference analysis

• Verification authority assessment

• Artifact modification recognition

• Artifact replacement analysis

• Promotion reference mapping

• Distribution reference mapping

• Version conflict detection

• Identity conflict detection

• Integrity uncertainty recognition

• Evidence gap detection

• Critical integrity gap detection

• Integrity failure recognition

• Integrity validation support

• Artifact continuity mapping

• Historical preservation analysis

• Artifact reassessment support

• Governed artifact-version integrity state representation

GIE™ may identify candidate version relationships, detect possible content mismatches, or propose artifact integrity assessments.

Such outputs shall remain distinguishable from sufficiently evidenced

integrity determinations.

GIE™ must not fabricate artifact provenance, invent integrity verification results, infer content identity from filenames, or represent a matching version label as proof of artifact integrity.

OOF® logic adapts to the governed reality. The engine adapts to OOF®

logic.

Governance Logic

The canonical AVIM governance logic is:

Receive Governed Configuration Baseline State

↓

Receive Governed Software Build State

↓

Identify Software Artifact Governance Object

↓

Establish Artifact Purpose, Type, Scope, and Classification

↓

Identify Artifact Identity

↓

Establish Artifact Content Identity Reference

↓

Identify Claimed Artifact Version

↓

Assess Version Attribution

↓

Assess Version Correspondence

↓

Identify Material Artifact Composition References

↓

Assess Configuration Baseline Correspondence

↓

Assess Build Correspondence

↓

Preserve Artifact Provenance References

↓

Identify Transformations and Derivations

↓

Assess Artifact Storage and Repository Records

↓

Identify Retrieved Artifact Content Where Applicable

↓

Associate Artifact Integrity Evidence

↓

Assess Verification Method Suitability

↓

Assess Integrity Reference Authenticity

↓

Identify Verification Authority

↓

Perform or Evaluate Competent Integrity Verification

↓

Identify Artifact Modifications and Replacements

↓

Identify Version and Identity Conflicts

↓

Identify Artifact Integrity Uncertainty

↓

Identify Evidence Gaps and Critical Integrity Gaps

↓

Identify Established Integrity Failures

↓

Determine Artifact Governance State

↓

Establish Governed Artifact Version Integrity State

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Preserve Artifact Continuity and Historical Relationships

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Handoff to Parent SCFS Governance

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Reassess When Material Artifact Conditions or Evidence Change

The principal output is:

Governed Artifact Version Integrity State

Governance Outputs

AVIM may produce:

Software Artifact Governance Object

Artifact Identity Record

Artifact Purpose Record

Artifact Type Record

Artifact Scope Record

Artifact Classification Record

Artifact Content Identity Record

Artifact Version Identity Record

Version Attribution Record

Version Correspondence Assessment

Artifact Composition Reference Record

Artifact Baseline Correspondence Record

Artifact Build Correspondence Record

Artifact Provenance Record

Artifact Transformation Record

Artifact Derivation Record

Artifact Equivalence Claim Record

Artifact Duplication Record

Artifact Storage Reference Record

Artifact Repository Record

Artifact Retrieval Correspondence Record

Artifact Integrity Evidence Register

Artifact Integrity Verification Record

Verification Method Suitability Assessment

Integrity Reference Authenticity Assessment

Artifact Signing Reference Record

Verification Authority Record

Artifact Modification Record

Artifact Replacement Record

Artifact Promotion Reference Record

Artifact Distribution Reference Record

Artifact Version Conflict

Artifact Identity Conflict

Artifact Integrity Uncertainty State

Artifact Evidence Gap

Critical Artifact Integrity Gap

Artifact Integrity Failure Record

Artifact Integrity Validation Record

Artifact Continuity Record

Artifact Historical Preservation Record

Artifact Reassessment Record

Artifact Governance State

SCFS Artifact Integrity Handoff Record

The principal output is:

Governed Artifact Version Integrity State

Critical Distinctions

AVIM preserves the following distinctions:

Declared Artifact Version ≠ Verified Artifact Version Integrity

Artifact Governance Object ≠ Artifact Filename

Artifact Purpose ≠ Artifact Authorization

Artifact Type ≠ Artifact Identity

Artifact Scope Definition ≠ Verified Artifact Composition

Artifact Classification ≠ Artifact Governance State

Artifact Name ≠ Verified Artifact Identity

Artifact Label ≠ Artifact Content Identity

Version Identifier ≠ Verified Artifact Version

Version Attribution Record ≠ Version Correspondence Evidence

Matching Version Label ≠ Verified Version Correspondence

Composite Artifact Identity ≠ Verified Constituent Integrity

Artifact Baseline Correspondence ≠ Verified Runtime Configuration

Build Reference ≠ Verified Artifact Build Origin

Recorded Artifact Provenance ≠ Verified Artifact Provenance

Artifact Transformation ≠ Artifact Identity Preservation

Verified Source Artifact ≠ Verified Derived Artifact

Functional Equivalence ≠ Content Identity

Duplicate Version Labels ≠ Identical Artifact Content

Artifact Storage Location ≠ Artifact Identity

Repository Record ≠ Independent Artifact Verification

Requested Artifact ≠ Retrieved Artifact

Integrity Evidence Presence ≠ Verified Artifact Integrity

Successful Integrity Check ≠ Complete Artifact Trustworthiness

Verification Mechanism Execution ≠ Competent Integrity Verification

Digest Match ≠ Trusted Artifact Origin

Valid Artifact Signature ≠ Authorized Software Release

Verification Capability ≠ Verification Authority

Previous Artifact Integrity ≠ Current Artifact Integrity

Artifact Replacement ≠ Artifact Identity Continuity

Artifact Promotion ≠ Release Authorization

Artifact Distribution ≠ Verified Artifact Receipt

Conflicting Version Records ≠ Permission to Select the Preferred Version

Shared Artifact Identifier ≠ Shared Artifact Identity

Unresolved Artifact Integrity ≠ Verified Artifact Integrity

Artifact Evidence Gap ≠ Proven Artifact Compromise

Critical Artifact Integrity Gap ≠ Automatic Artifact Compromise

Verified Integrity Failure ≠ Mere Integrity Uncertainty

Validated Artifact Integrity ≠ Validated Software Correctness

Artifact Continuity ≠ Artifact Content Immutability

Historical Artifact Integrity ≠ Current Artifact Suitability

Artifact Reassessment ≠ Historical Integrity Record Erasure

Verified Artifact Integrity ≠ Verified Software Safety

Cross-Architecture Boundary

AVIM governs software artifact identity, version correspondence, content

integrity, provenance references, and lifecycle continuity within SCFS and SLGA™.

It does not independently establish configuration baselines, execute builds, authorize configuration changes, demonstrate build reproducibility, validate software quality, certify software security, authorize release, or authorize deployment.

Within SCFS:

CBM — Configuration Baseline Module governs configuration baseline

identity, authority, composition, applicability, evidence, and continuity.

BMM — Build Management Module governs build identity, inputs, tools,

parameters, authority, execution, observation, and resulting-artifact correspondence.

AVIM — Artifact Version Integrity Module governs artifact identity,

content identity, version attribution, version correspondence, integrity evidence, and lifecycle continuity.

CCCM — Configuration Change Control Module governs configuration change

authority, execution, and resulting-state relationships.

BREM — Build Reproducibility & Environment Module governs build

environment identity, reproducibility conditions, and reconstruction evidence.

Within SLGA™:

Software Requirements Standard (SRS) governs software requirements.

Software Creation Standard (SCS) governs software creation.

Software Dependency Standard (SDS) governs software dependencies.

Software Quality Standard (SQS) governs software quality.

Software Security Standard (SSS) governs software security.

Software Release Standard (SRLS) governs software release.

Software Deployment Standard (SDPS) governs software deployment.

Software Change Management Standard (SCMS) governs broader lifecycle

change management.

Software Retirement Standard (SRTS) governs software retirement.

Software Lifecycle Control Standard (SLCS) governs cross-cutting

lifecycle control.

Relevant interoperating OOF® architectures include:

GOA™ — Governance Architecture: General authority and delegation

governance.

AGA™ — Accountability Governance Architecture: Relevant responsibility

and accountability relationships.

ORA™ — Operational Reality Architecture: Competent assessment of actual

artifact and operational conditions.

ASGA™ — Autonomous Systems Governance Architecture: Relevant autonomous

artifact handling and operational boundaries.

AGI™ — Agent Governance Infrastructure: Relevant agent permissions and

tool-path governance.

AVIM preserves artifact-version-specific governance without replacing these

competent domains.

Canonical Boundary

AVIM begins when a material software artifact requires identifiable content, version attribution, integrity verification, provenance assessment, or lifecycle continuity governance.

AVIM governs:

- Artifact governance objects
- Artifact purpose
- Artifact type
- Artifact scope
- Artifact classification
- Artifact identity
- Artifact content identity
- Artifact version identity
- Version attribution
- Version correspondence
- Artifact composition references
- Configuration baseline correspondence
- Build correspondence
- Artifact provenance references
- Artifact transformations
- Artifact derivation
- Artifact equivalence claims
- Artifact duplication
- Artifact storage references
- Repository records
- Artifact retrieval correspondence
- Integrity evidence
- Integrity verification
- Verification method suitability
- Integrity reference authenticity
- Artifact signing references
- Verification authority

• Artifact modification recognition

• Artifact replacement

• Artifact promotion references

• Artifact distribution references

• Artifact version conflicts

• Artifact identity conflicts

• Artifact integrity uncertainty

• Artifact evidence gaps

• Critical artifact integrity gaps

• Artifact integrity failures

• Artifact integrity validation

• Artifact continuity

• Historical preservation

• Artifact reassessment

• Artifact governance states

AVIM does not independently determine:

• Software requirements validity

• Software creation authorization

• Complete dependency suitability

• Configuration baseline establishment

• Build execution correctness

• Build reproducibility

• Software quality

• Software security

• Release authorization

• Deployment authorization

• Legal compliance

• Legal liability

AVIM ends when the material artifact-version integrity conditions have been represented as a:

Governed Artifact Version Integrity State

with sufficient artifact identity, content, version, provenance, build

correspondence, integrity evidence, validation, uncertainty, and continuity information for competent parent SCFS governance.

Use Case 1 — Two Artifacts Share the Same Version Label but Contain

Different Software

An organization maintains a software application distributed through two internal artifact repositories.

Both repositories contain an artifact labeled:

Application Version 4.2.1

The first artifact was generated through an approved build pipeline.

The second artifact was uploaded later by an automated maintenance process.

Both artifacts have the same filename and version label.

However, their cryptographic digests differ.

The organization initially assumes that the artifacts are interchangeable because the version labels match.

AVIM evaluates:

Artifact Identity → Version Attribution → Content Identity → Build

Correspondence → Repository Evidence → Integrity Verification

The assessment identifies:

• Two distinct artifact content identities

• One shared version label

• Conflicting version correspondence claims

• Different artifact creation or registration histories

• Insufficient evidence that both artifacts represent the same

authorized version.

The relevant governance state may be:

Conflicted

until competent evidence resolves the version correspondence.

The critical distinctions are:

Matching Version Label ≠ Verified Version Correspondence.

Duplicate Version Labels ≠ Identical Artifact Content.

Repository Record ≠ Independent Artifact Verification.

Artifact Name ≠ Verified Artifact Identity.

The result: AVIM prevents version labels from concealing materially

different software artifacts and establishes the need for content-based identity and evidence-supported version correspondence.

Use Case 2 — A Software Package Passes a Checksum Test Against an

Untrusted Reference

An organization downloads a software package from an external distribution

source.

The package is accompanied by a checksum published on the same website.

The organization calculates the package checksum.

The values match.

The package is therefore initially considered verified.

However, the distribution website has been compromised.

Both the software package and the published checksum were replaced by the

attacker.

The checksum match is technically correct.

But the reference itself is not trustworthy.

AVIM evaluates:

Retrieved Artifact → Content Identity → Integrity Reference → Reference

Authenticity → Provenance → Verification Evidence

The assessment identifies:

• The downloaded package matches the supplied checksum

• The checksum originates from the same compromised distribution source

• The match supports content correspondence with the supplied reference

• The match does not establish that the reference represents the

• Independent provenance or trusted reference evidence is required for

intended authorized artifact. the broader integrity claim.

The relevant governance state may be:

Provenance-Unresolved

or:

Integrity-Unresolved

depending on the available evidence.

The critical distinctions are:

Digest Match ≠ Trusted Artifact Origin.

Integrity Evidence Presence ≠ Verified Artifact Integrity.

Verification Mechanism Execution ≠ Competent Integrity Verification.

Successful Integrity Check ≠ Complete Artifact Trustworthiness.

The result: AVIM prevents technically successful integrity checks from

being treated as sufficient proof when the underlying integrity reference cannot be trusted.

Canonical Closing Statement

Software artifacts are not reliably identified by their names.

They are not reliably verified by version labels alone.

And they do not become trustworthy merely because they exist in a

repository, carry a digital signature, or match a supplied checksum.

An artifact's identity must remain connected to its actual content.

Its version must remain connected to evidence supporting version correspondence.

Its provenance must remain distinguishable from an unsupported origin claim.

Its integrity verification must rely on competent methods and sufficiently trustworthy references.

And its lifecycle history must preserve the material relationships between original, transformed, copied, replaced, and retrieved artifacts.

Declared Artifact Version ≠ Verified Artifact Version Integrity.

Matching Version Label ≠ Verified Version Correspondence.

Recorded Artifact Provenance ≠ Verified Artifact Provenance.

Digest Match ≠ Trusted Artifact Origin.

Validated Artifact Integrity ≠ Validated Software Correctness.

These distinctions become increasingly important as software artifacts are generated, transformed, stored, and distributed through automated systems and autonomous software development processes.

An AI agent may create a package.

A build pipeline may assign a version.

A repository may register the artifact.

A distribution system may deliver it.

None of these actions independently establishes that the artifact's claimed identity, content, version, and provenance remain sufficiently supported.

AVIM establishes the methodological requirement that artifact-version integrity remain an identifiable, evidence-supported governance condition rather than an assumption derived from filenames, labels, or successful technical operations.

It does not promise that an integrity-verified artifact is secure, functionally correct, or authorized for release.

It establishes the narrower but essential condition that the artifact under assessment is sufficiently identifiable and that its material version and integrity claims can be competently evaluated.

AVIM transforms software artifact versions from administrative labels into

governed identity and integrity relationships — ensuring that software organizations can establish not merely which version an artifact claims to represent, but which artifact actually exists, what content it contains, how its version is supported, and whether its integrity remains verifiable throughout the software lifecycle.

An artifact version is not governed because it has been named, numbered, signed, or stored. It is governed when its identity, content, version correspondence, provenance, integrity evidence, and lifecycle continuity remain sufficiently established to support competent software configuration governance.